Entity Name: Member

Property Name	Data Type	Source Tag
memberID	int	System Generated
memberName	string	User Input
PhoneNumber	string	User Input
memberEmail	string	User Input
registrationDate	string	Calculated

Entity Name: Book

Property Name	Data Type	Source Tag
bookID	int	System Generated
bookTitle	String	User Input
book Author	String	User Input
bookGenre	String	User Input
book Copies	int	User Input
isAvailableCopy	bool	Default Value

Entity Name: Loan

Property Name	Data Type	Source Tag
LoanID	int	System Generated
borrowDate	string	System Generated
return Date	String	System Generated
member ID	int	From list
book ID	int	From list
due Date	String	Calculated
fineAmount	decimal	Calculated

Entity Name: Reservation

Property Name	Data Type	Source Tag
reservation ID	int	System Generated
member ID	int	From List
book ID	int	From List
reservationDate	string	System Generated
notificationMessage	bool	Default value

If you identify more than 4 entities, continue on the back of this sheet or add extra rows above.

Step 2 — Describe the Entity Relationships (Phase 1 continued)

How do the entities connect to each other? Describe the full cycle in plain English — who does what, which entity gets created, and what changes in other entities as a result.

Write the relationship cycle here:

- Member registers in library.
- Librarian Adds books to the System.
- Member borrow or reserve books.
- Member return book before 14 days, if late calculate a fine.
- When a copy of book available, the system notifies the reserved member

Step 3 — List the Services (Phase 4)

From the description, identify every service (function) the system needs. Write at least 8 services — give each a name and a one-line description of what it does.

#	Service Name	What it does
1	RegisterMember	Add, List (Members)
2	AddBooks	Add, List(Books)
3	BorrowBook	read, internal read, List(Books, loan)
4	ReturnBook	Update, internal read
5	ReserveBook	Add, internal read, list(Book, Reservation)
6	viewActiveLoans	read, list (loan)
7	ViewOverdueLoans	read, internal read
8	CalculateLateFine	read, list(loan)
9	MostBorrowBookReport	read, list(loan)
10		

Domain 2 — Hotel Management System

Assign the source tags

The class models and all their properties are already defined for you below. Your job is to **fill in the Source Tag column** for every single property. Use the five source tags from the legend on page 1: System Generated, User Input, Default Value, Calculated, From List.

A hotel allows guests to register and book rooms for a stay. The hotel has different room types (Single, Double, Suite) each with its own nightly price. Each room can only have one active booking at a time. When a guest checks in, a check-in record is created. The total price of a stay is calculated from the number of nights and the room's nightly rate. When a guest checks out, the room becomes available again. A booking can also be cancelled before check-in, which frees the room immediately.

Your task: For each property in the tables below, write the correct source tag in the empty column on the right. Think carefully — ask yourself: does the user type this? does the system generate it? is it copied from another object? does it have a fixed starting value? does the user pick it from a list?

Guest

Property Name	Data Type	Source Tag (fill in)
`guestId`	int	System Generated
`guestName`	string	User Input
`guestPhone`	string	User Input
`guestEmail`	string	User Input
`guestNationality`	string	User Input
`registrationDate`	string	User Input

Room

Property Name	Data Type	Source Tag (fill in)
roomId	int	System Generated
roomNumber	string	User Input
roomType	string	User Input
floorNumber	int	User Input
nightlyPrice	decimal	User Input
isAvailable	bool	Default Value

Booking

Property Name	Data Type	Source Tag (fill in)
bookingId	int	System Generated
guestId	int	From List
roomId	int	From List
checkInDate	string	User Input
checkOutDate	string	User Input
numberOfNights	int	Calculated
totalPrice	decimal	Calculated
status	string	Default value

CheckInRecord		
Property Name	**Data Type**	**Source Tag (fill in)**
recordId	int	System Generated
bookingId	int	From List
guestId	int	Calculated
roomId	int	Calculated
actualCheckIn	string	System Generated
actualCheckOut	string	System Generated
finalAmount	decimal	Calculated

Step 2 — Describe the Entity Relationships

Now that you have studied the four class models, describe in your own words how they connect to each other. Who creates what? What changes when a booking is made? What happens when a guest checks out?

Write the relationship cycle here:

- Guest regrsters in the hotel.
- Rooms are added and managed by hotel.
- Guest Select an available room and creates a booking.
- Number of nights calculated by check-in and check-out dates.
- ~~Total price calculated using nights * nightly Price~~
- After check-out, the room become available.

Step 3 — List the Services

Based on the description and the class models above, list all the services (functions) this hotel system needs. Give each a name and a one-line description.

#	Service Name	What it does
1	Register Guest	Add, list (Guest)
2	Add rooms	Add, list (Room)
3	ViewAvailableRooms	read, list (Room)
4	CreateBooking	Add, internal read, list (Booking, Room)
5	cancelBooking	update, internal read, list (Booking, Room)
6	Check In Guest	read, internal read, list (Booking, CheckInRe)
7	Check out Guest	read, internal read, list (Booking, Check inRe

#	Service Name	What it does
8	Calculate Total Price	read , list (Booking), Room)
9	view Active Booking	read , list (Booking)